Description:**Spotnana Technical Assessment 2026**

Start

Enhanced Data Setup

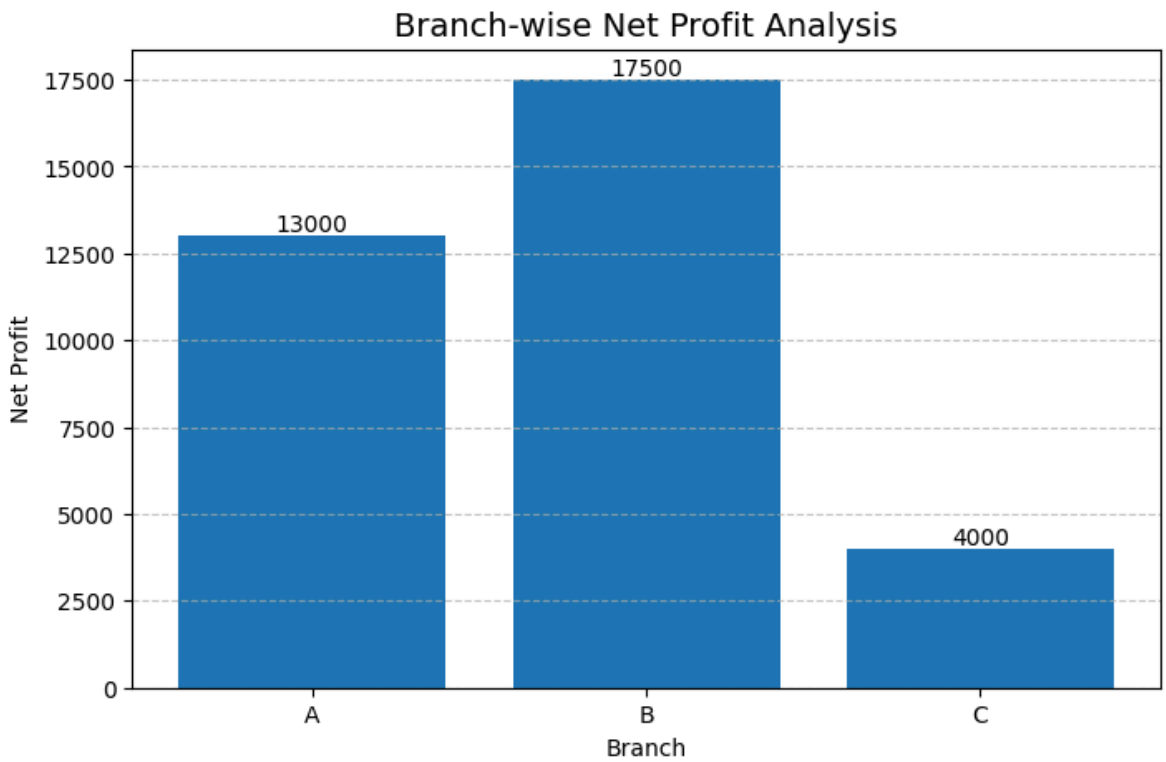
```
import pandas as pd
import numpy as np


data = {
    "Branch": ["A", "B", "C"],
    "Total Sales": [45000, 38500, 52000],
    "Operating Cost": [32000, 21000, 48000],
    "Customer Feedback": [
        "Great service, but long queues.",
        "Always has stock, friendly staff.",
        "Expensive shipping, slow delivery."
    ]
}

df = pd.DataFrame(data)

# Calculate Net Profit

df["Net Profit"] = df["Total Sales"] - df["Operating Cost"]

# Calculate Profit Margin %

df["Profit Margin (%)"] = (df["Net Profit"] / df["Total Sales"]) * 100

df
```

	Branch	Total Sales	Operating Cost	Customer Feedback	Net Profit	Profit Margin (%)
0	A	45000	32000	Great service, but long queues.	13000	28.888889
1	B	38500	21000	Always has stock, friendly staff.	17500	45.454545
2	C	52000	48000	Expensive shipping, slow delivery.	4000	7.692308

Next steps: Generate code with df New interactive sheet

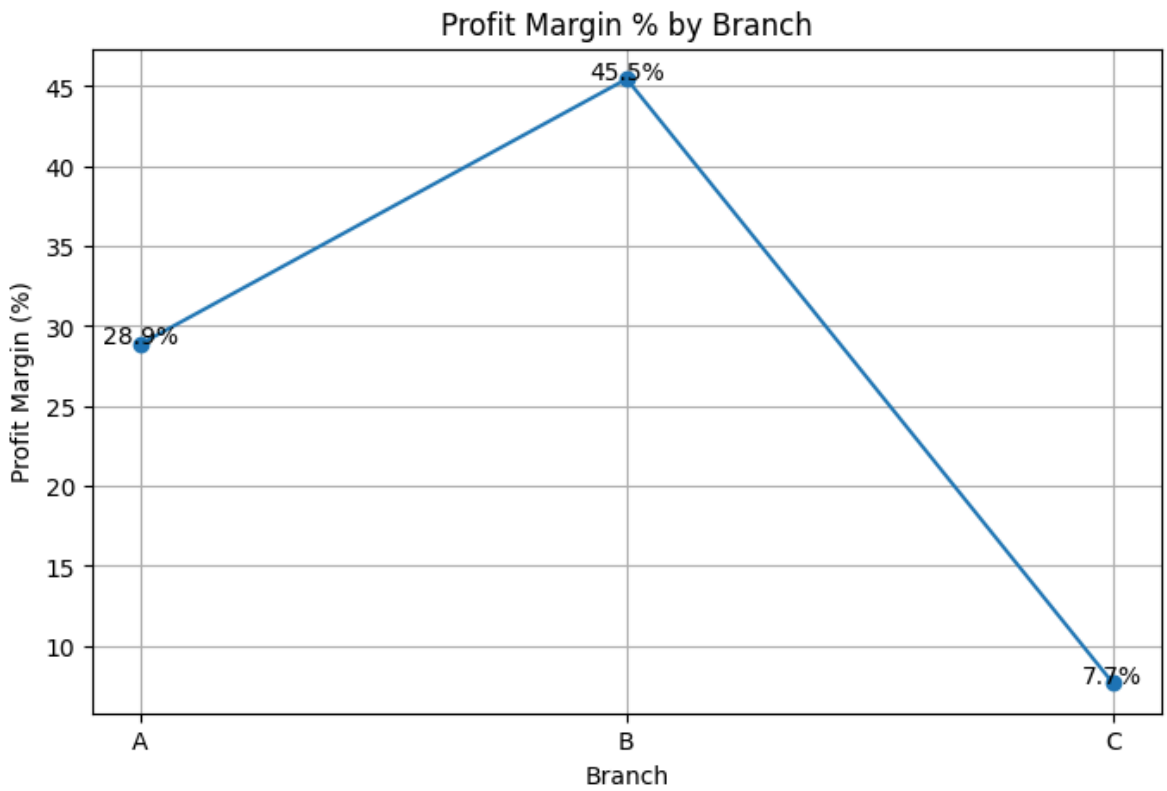
Advanced Visualization

```
import matplotlib.pyplot as plt


plt.figure(figsize=(8,5))

bars = plt.bar(df["Branch"], df["Net Profit"])

for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height(),
             int(bar.get_height()),
             ha='center', va='bottom')


plt.title("Branch-wise Net Profit Analysis", fontsize=14)

plt.xlabel("Branch")

plt.ylabel("Net Profit")

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```



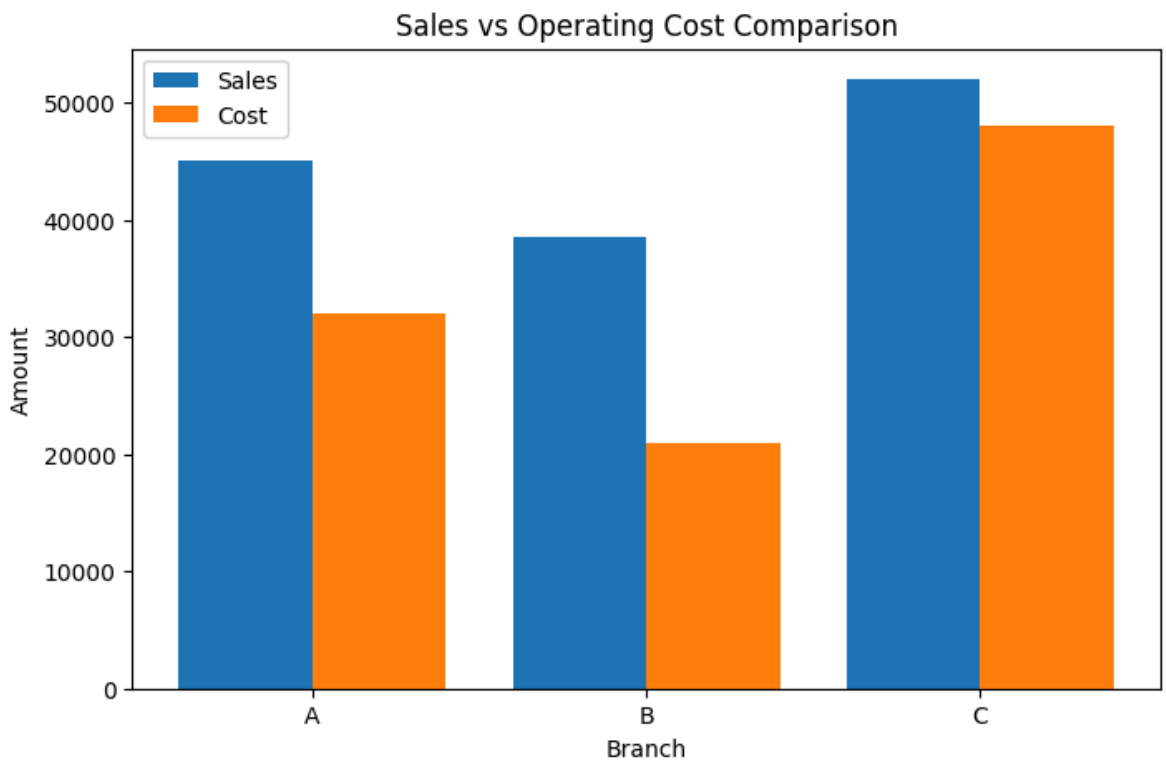
Profit Margin Comparison (Insight Chart)

```
plt.figure(figsize=(8,5))
plt.plot(df["Branch"], df["Profit Margin (%)"], marker='o')
for i, val in enumerate(df["Profit Margin (%)"]):
    plt.text(i, val, f"{val:.1f}%", ha='center')
plt.title("Profit Margin % by Branch")
plt.xlabel("Branch")
plt.ylabel("Profit Margin (%)")
plt.grid()
plt.show()
```



Sales vs Cost (Business Insight Chart)

```
x = np.arange(len(df["Branch"]))
plt.figure(figsize=(8,5))
plt.bar(x - 0.2, df["Total Sales"], width=0.4, label="Sales")
plt.bar(x + 0.2, df["Operating Cost"], width=0.4, label="Cost")
plt.xticks(x, df["Branch"])
plt.title("Sales vs Operating Cost Comparison")
plt.xlabel("Branch")
plt.ylabel("Amount")
plt.legend()
plt.show()
```



Smart Sentiment Analysis

```
from textblob import TextBlob
# Function to get sentiment polarity
def get_sentiment(text):
    return TextBlob(text).sentiment.polarity
df["Sentiment Score"] = df["Customer Feedback"].apply(get_sentiment)
# Rank (higher score = better experience)
df["Customer Rank"] = df["Sentiment Score"].rank(ascending=False)
df.sort_values("Customer Rank")
```

	Branch	Total Sales	Operating Cost	Customer Feedback	Net Profit	Profit Margin (%)	Sentiment Score	Customer Rank
0	A	45000	32000	Great service, but long queues.	13000	28.888889	0.375	1.5
1	B	38500	21000	Always has stock, friendly staff.	17500	45.454545	0.375	1.5
2	C	52000	48000	Expensive shipping, slow delivery.	4000	7.692308	-0.400	3.0

```
print("📊 BUSINESS KPI SUMMARY")
print("-"*30)
for i in range(len(df)):
    print(f"Branch {df['Branch'][i]}:")
    print(f" Net Profit: {df['Net Profit'][i]}")
    print(f" Profit Margin: {df['Profit Margin (%)'][i]:.2f}%")
    print(f" Sentiment Score: {df['Sentiment Score'][i]:.2f}")
    print("-"*30)
```

```
📊 BUSINESS KPI SUMMARY
------------------------------
Branch A:
 Net Profit: 13000
 Profit Margin: 28.89%
 Sentiment Score: 0.38
------------------------------
Branch B:
 Net Profit: 17500
 Profit Margin: 45.45%
 Sentiment Score: 0.38
------------------------------
Branch C:
 Net Profit: 4000
 Profit Margin: 7.69%
 Sentiment Score: -0.40
------------------------------
```

Bonus Business Recommendation

Branch B should receive the Performance Bonus as it demonstrates the highest profitability and excellent customer satisfaction, indicating strong operational efficiency. Branch A performs moderately well but requires process improvements to reduce customer wait times and enhance service flow. Branch C, despite having the highest sales, suffers from very low profitability due to high operating costs and poor customer feedback. Therefore, Branch C should be prioritized for a Process Improvement plan focusing on cost optimization and delivery efficiency.

```
# Reduce Branch C cost by 15%
branch_c = df[df["Branch"] == "C"].copy()
branch_c["New Cost"] = branch_c["Operating Cost"] * 0.85
branch_c["New Net Profit"] = branch_c["Total Sales"] - branch_c["New Cost"]
branch_c[["Branch", "New Cost", "New Net Profit"]]
```

	Branch	New Cost	New Net Profit
2	C	40800.0	11200.0

Scenario Analysis (Cost Optimization)

```
# Scenario: Reduce Branch C cost by 15%
df["New Cost"] = df["Operating Cost"]
df.loc[df["Branch"] == "C", "New Cost"] *= 0.85
df["New Profit"] = df["Total Sales"] - df["New Cost"]
df[["Branch", "New Cost", "New Profit"]]
```

	Branch	New Cost	New Profit
0	A	32000	13000
1	B	21000	17500
2	C	40800	11200

The analysis shows that Branch B is the best performer due to high profitability and strong customer satisfaction. Branch A performs steadily but needs operational improvements, while Branch C, despite high sales, suffers from low profitability due to high costs. Overall, the study highlights that efficiency and cost control are key drivers of business success.

Best possible key Outcome

1. Branch B ⟶ Best performance (high profit + good feedback)
2. Branch A ⟶ Average performance (needs process improvement)
3. Branch C ⟶ Poor profitability (high cost issue)
4. Customer ranking: B > A > C
5. Cost efficiency matters more than sales
6. Branch C profit improves with cost reduction
7. Bonus ⟶ Branch B
8. Improvement ⟶ Branch C (priority), Branch A (secondary)

Thankyou